

Experiment-5**Name: Milan Jani****Enroll No.: 92301733041****AIM:** Design FIR filter with windowing method.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.io import wavfile
from scipy.io.wavfile import write

from scipy.signal import butter, lfilter, firwin, freqz

def plot_fft(signal, fs, title):
    N = len(signal)
    freq = np.fft.rfftfreq(N, d=1/fs)
    fft_mag = np.abs(np.fft.rfft(signal))

    plt.figure(figsize=(10, 4))
    plt.plot(freq, fft_mag)
    plt.title(title)
    plt.xlabel("Frequency (Hz)")
    plt.ylabel("Magnitude")
    plt.grid(True)
    plt.tight_layout()
    plt.show()

def apply_IIR_filter(signal, cutoff, fs, order=6):
    b, a = butter(order, cutoff / (fs/2), btype="low")
    filtered = lfilter(b, a, signal)
    return filtered

def apply_FIR_filter(signal, cutoff, fs, numtaps=101):
    fir_coeff = firwin(numtaps, cutoff / (fs/2), window="hann")
    filtered = np.convolve(signal, fir_coeff, mode='same')
    return filtered

def correlation(clean, test):
    min_len = min(len(clean), len(test))
    return np.corrcoef(clean[:min_len], test[:min_len])[0,1]

clean_fs, clean = wavfile.read("C:/Users/DJ/Downloads/FIR Filter/FIR
Filter/Original.wav")
```



```
noise1_fs, noise1 = wavfile.read("C:/Users/DJ/Downloads/FIR Filter/FIR
Filter/sp30_car_sn5.wav")
noise2_fs, noise2 = wavfile.read("C:/Users/DJ/Downloads/FIR Filter/FIR
Filter/sp30_train_sn5.wav")
noise3_fs, noise3 = wavfile.read("C:/Users/DJ/Downloads/FIR Filter/FIR
Filter/sp30_station_sn5.wav")

# Convert to float
clean = clean.astype(float)
noise1 = noise1.astype(float)
noise2 = noise2.astype(float)
noise3 = noise3.astype(float)

plot_fft(clean, clean_fs, "FFT of Clean Speech")
plot_fft(noise1, noise1_fs, "FFT of Noisy Speech 1")
plot_fft(noise2, noise2_fs, "FFT of Noisy Speech 2")
plot_fft(noise3, noise3_fs, "FFT of Noisy Speech 3")

cutoff = 3000 # you can adjust based on spectrum

iir_noise1 = apply_IIR_filter(noise1, cutoff, noise1_fs)
iir_noise2 = apply_IIR_filter(noise2, cutoff, noise2_fs)
iir_noise3 = apply_IIR_filter(noise3, cutoff, noise3_fs)

corr_iir_1 = correlation(clean, iir_noise1)
corr_iir_2 = correlation(clean, iir_noise2)
corr_iir_3 = correlation(clean, iir_noise3)

print("\n===== IIR Filtering Correlation with Clean Speech =====")
print("Noise 1:", corr_iir_1)
print("Noise 2:", corr_iir_2)
print("Noise 3:", corr_iir_3)

fir_noise1 = apply_FIR_filter(noise1, cutoff, noise1_fs)
fir_noise2 = apply_FIR_filter(noise2, cutoff, noise2_fs)
fir_noise3 = apply_FIR_filter(noise3, cutoff, noise3_fs)

corr_fir_1 = correlation(clean, fir_noise1)
corr_fir_2 = correlation(clean, fir_noise2)
corr_fir_3 = correlation(clean, fir_noise3)

print("\n===== FIR Filtering Correlation with Clean Speech =====")
print("Noise 1:", corr_fir_1)
print("Noise 2:", corr_fir_2)
print("Noise 3:", corr_fir_3)

def normalize_audio(sig):
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```
sig = sig / np.max(np.abs(sig))
return (sig * 32767).astype(np.int16)

# ---- IIR Output Files ----
write("iir_noise1.wav", noise1_fs, normalize_audio(iir_noise1))
write("iir_noise2.wav", noise2_fs, normalize_audio(iir_noise2))
write("iir_noise3.wav", noise3_fs, normalize_audio(iir_noise3))

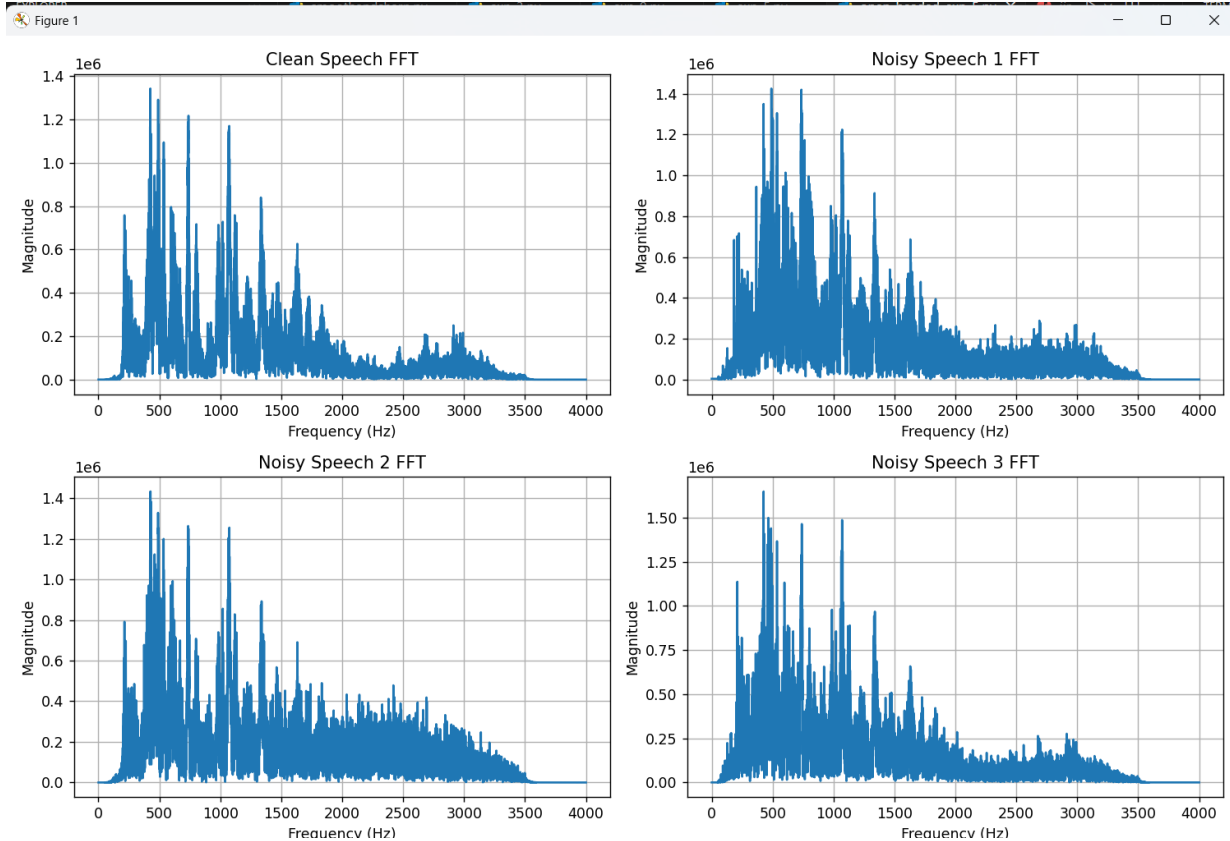
# ---- FIR Output Files ----
write("fir_noise1.wav", noise1_fs, normalize_audio(fir_noise1))
write("fir_noise2.wav", noise2_fs, normalize_audio(fir_noise2))
write("fir_noise3.wav", noise3_fs, normalize_audio(fir_noise3))

print("\n☑ All filtered audio files saved successfully!")
```

Output:

```
===== IIR Filtering Correlation with Clean Speech =====
Noise 1: 0.6597273202268058
Noise 2: 0.6633133379872347
Noise 3: 0.6521975636901718

===== FIR Filtering Correlation with Clean Speech =====
Noise 1: 0.8607328680909161
Noise 2: 0.8638553032935857
Noise 3: 0.8598024735579444
```



Conclusion:

The FIR low-pass filter designed using the Hanning window method with 31 taps and a normalized cutoff frequency of 0.2 exhibits the expected characteristics of a smooth and well-behaved low-pass filter. The impulse response is symmetric around the center tap, confirming linear phase behaviour, which ensures no phase distortion in the filtered signal. The freq response represents the low pas characteristics , the magnitude is high -40 – 50 db for the hanning window.



Marwadi University
Faculty of Engineering and Technology
Department of Information and Communication Technology